



Experiment 3

Experiment Title

Implement Network Address Translation (NAT) for Public-Private Network Communication.

Problem Statement

Design and configure a network using Cisco Packet Tracer consisting of a private LAN connected to a public network through a router. Implement Static Network Address Translation (Static NAT) to establish one-to-one mapping between a private IP address and a public IP address. Verify successful communication between the private and public networks using NAT translation tables and connectivity tests.

Theory

Introduction to Network Address Translation (NAT)

Network Address Translation (NAT) is a networking technique used to translate private IP addresses into public IP addresses, allowing devices in a private network to communicate with external networks such as the Internet. NAT is configured on the boundary router that connects the internal network to the external network.

Static NAT provides a permanent one-to-one mapping between a private IP address and a public IP address. Whenever packets are sent from the private host, the router replaces the private IP address with the configured public IP address. Similarly, packets arriving from the public network are translated back to the private IP address before forwarding them to the internal host. Static NAT is mainly used for servers that require a fixed public IP address, such as Web, FTP, and Mail servers.

Features

- Permanent one-to-one IP address mapping
- Supports both inbound and outbound communication
- Easy to configure and manage
- Provides fixed public IP address for internal hosts
- Translation entry remains active until removed

Advantages

- ✓ Simple implementation
- ✓ Allows external access to internal servers
- ✓ Predictable IP address mapping
- ✓ Improves security by hiding private addresses

Limitations

- Requires one public IP address for every private host
- Does not conserve public IP addresses
- Not suitable for large-scale networks

Applications

- Web Servers
- Mail Servers

- Enterprise Networks
- Educational Laboratory Experiments

Steps

1. Open Cisco Packet Tracer.
2. Create the required network topology.
3. Connect the router, switches, PC, and server.
4. Assign IP addresses to all devices.
5. Configure the router interfaces as Inside and Outside interfaces.
6. Configure Static NAT mapping.
7. Verify NAT translation.
8. Test end-to-end connectivity using Ping.
9. Save the project.

Commands / Procedure

Network Topology

Devices Required

One Router (Cisco 2911)

Two Switches (2960)

One PC

One Server

Copper Straight-through Cables

IP Addressing		
Device	Interface	IP Address
PC1	NIC	192.168.1.2/24
Router	G0/0 (Inside)	192.168.1.1/24
Router	G0/1 (Outside)	200.1.1.1/24
Server	NIC	200.1.1.2/24

Step 1: Configure Router Interfaces

```
Router> enable
Router# configure terminal
Router(config)# interface g0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
```

```
Router(config)# interface g0/1
Router(config-if)# ip address 200.1.1.1 255.255.255.0
Router(config-if)# ip nat outside
Router(config-if)# no shutdown
```

Step 2: Configure Static NAT

```
Router(config)# ip nat inside source static 192.168.1.2 200.1.1.10
```

Step 3: Verify NAT Configuration

```
Router# show ip nat translations
Router# show ip nat statistics
Router# show running-config
```

NAT Table for Router1				
Protocol	Inside Global	Inside Local	Outside Local	Outside Global
----	30.30.30.1	10.10.10.1	----	----

Step 4: Test Connectivity

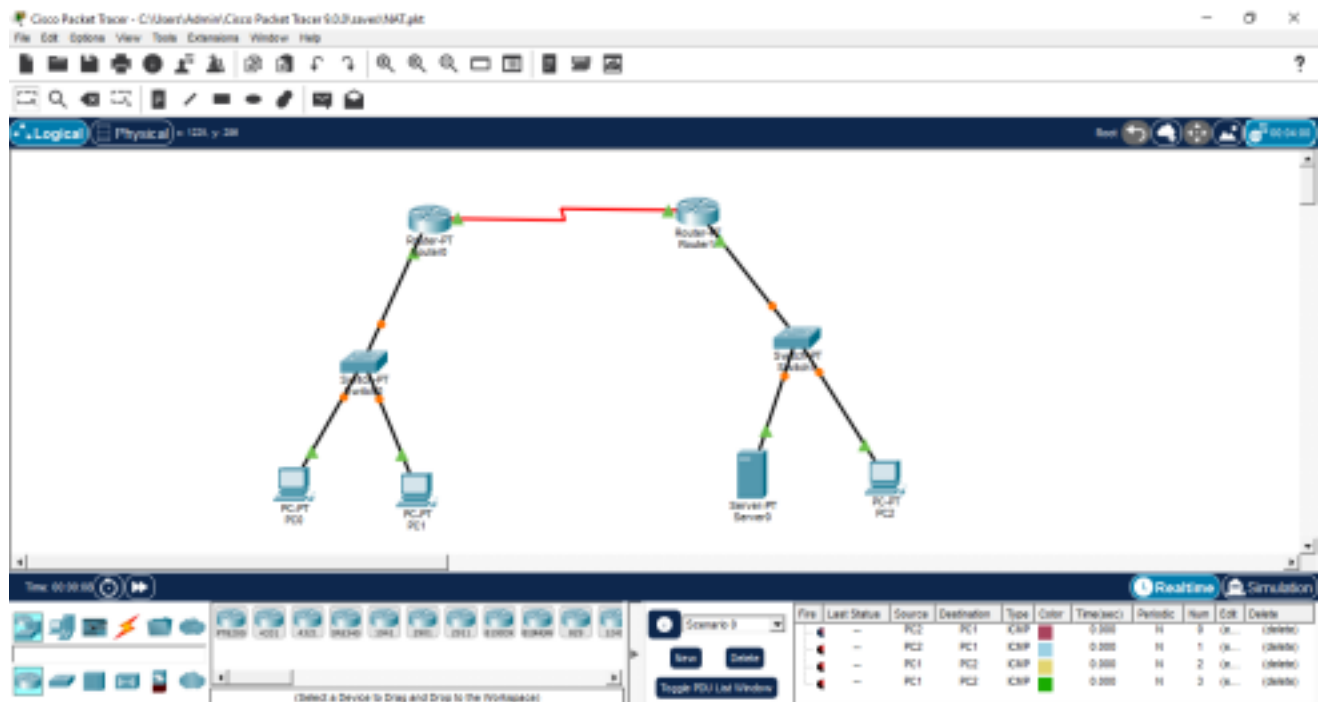
From PC1
ping 200.1.1.2

*** NAT Translation



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show ip nat translations



Result

Successfully configured Static Network Address Translation (Static NAT) to establish communication between the private and public networks. Verified the NAT translation using the NAT translation table and successful ping responses.

Conclusion

Successfully implemented Static Network Address Translation (NAT) in Cisco Packet Tracer by configuring a permanent one-to-one mapping between a private IP address and a public IP address. Verified successful communication between the private and public networks using NAT translation entries and connectivity tests.